

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2 Answer Book

NAME: _____ *Memo*

CLASS: 12 _____ TEACHER: _____

Question	Mark			Marker	HOD
	H	T	U		
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
TOTAL					
%					

QUESTION 1

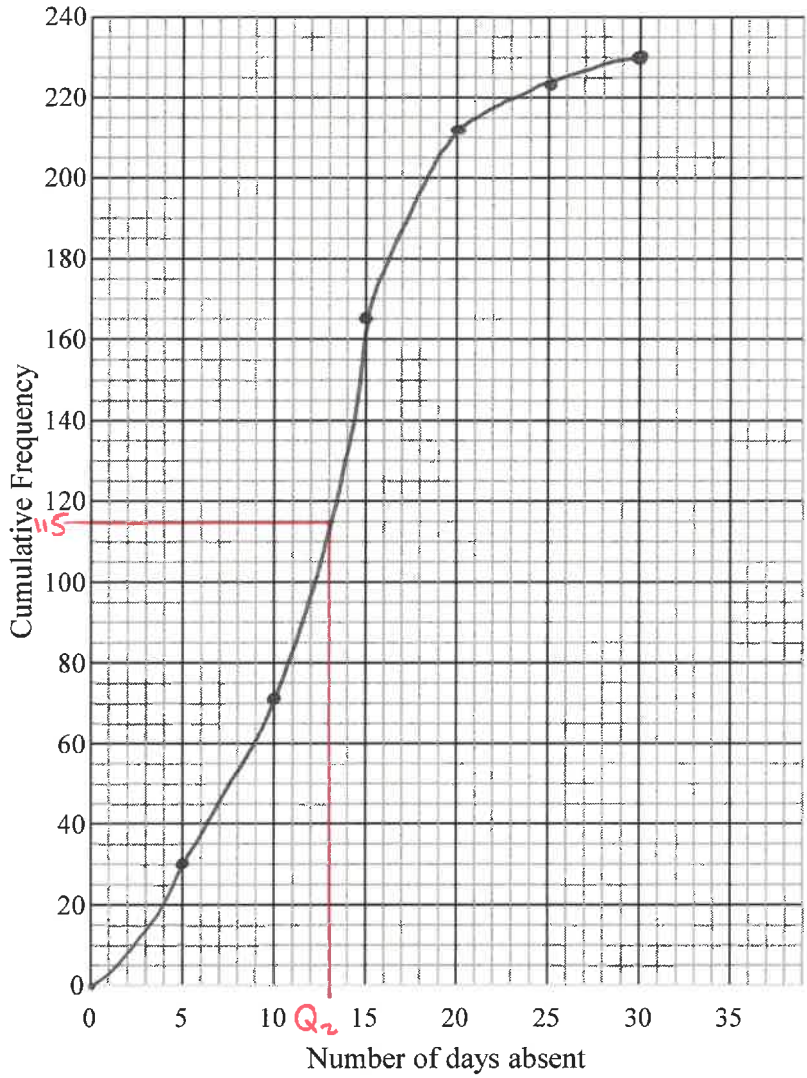
17	69	10	58	34	18	13	26	3	12	24	16	15	40	20
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	Solution	Marks
1.1.1	$\sigma = 17,65$ ✓	(1)
1.1.2	$\bar{x} = 25$ ✓	(1)
1.2	$\bar{x} + \sigma = 42,65$ ✓ $\therefore 2 \text{ days}$ ✓	(2)
1.3	$\frac{375 + x}{30} = 20$ ✓ $\therefore x = 225 \text{ MB}$ ✓	(2)
		[6]

QUESTION 2

Number of days absent	Number of learners	CF
$0 \leq x < 5$	30	30
$5 \leq x < 10$	41	71
$10 \leq x < 15$	94	165
$15 \leq x < 20$	47	212
$20 \leq x < 25$	11	223
$25 \leq x < 30$	7	230

	Solution	Marks
2.1	$10 \leq x < 15$ ✓	(1)
2.2	165 ✓	(1)
2.3	230 ✓	(1)

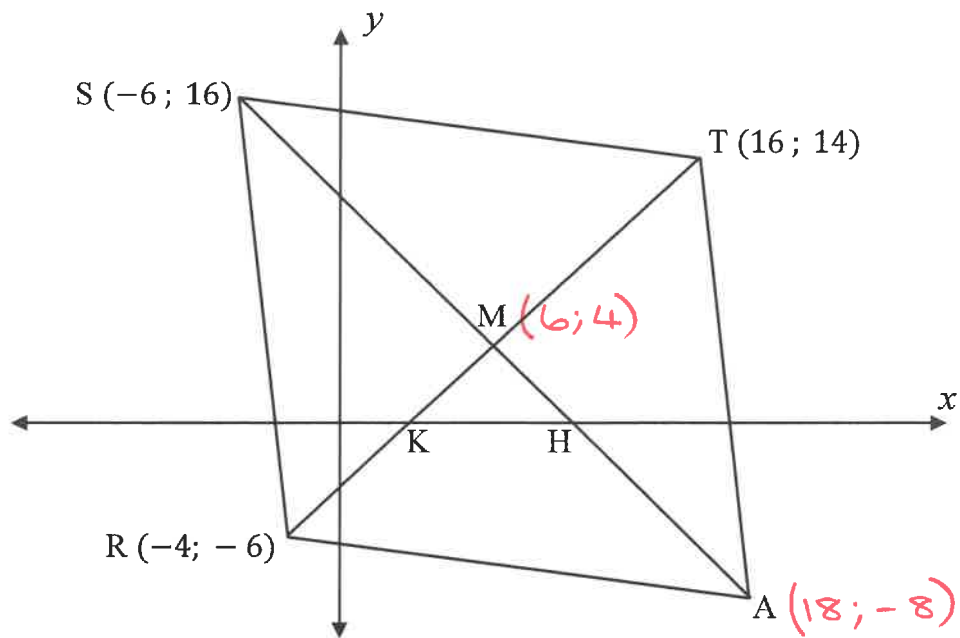
	Solution	Marks
2.4	 Cumulative Frequency Number of days absent Q_2	✓✓✓ -1 inaccuracy. (4)
2.5	13 days ✓ ±1 day ✓ in on graph	(2) [9]

QUESTION 3

Wind Speed in km/h (x)	13	11	25	15	2	6	20	17	24	22
Temperature in °C (y)	19	19	26	24	16	22	28	26	22	20

	Solution	Marks
3.1	$\hat{y} = a + bx$ $\hat{y} = 17,52 + 0,30x$	(2)
3.2	$\hat{y} = 17,52 + 0,30(9)$ $\hat{y} = 20,22^{\circ}\text{C}$	(2)
3.3	$r = 0,61$ Moderate positive correlation	(2)
		[6]

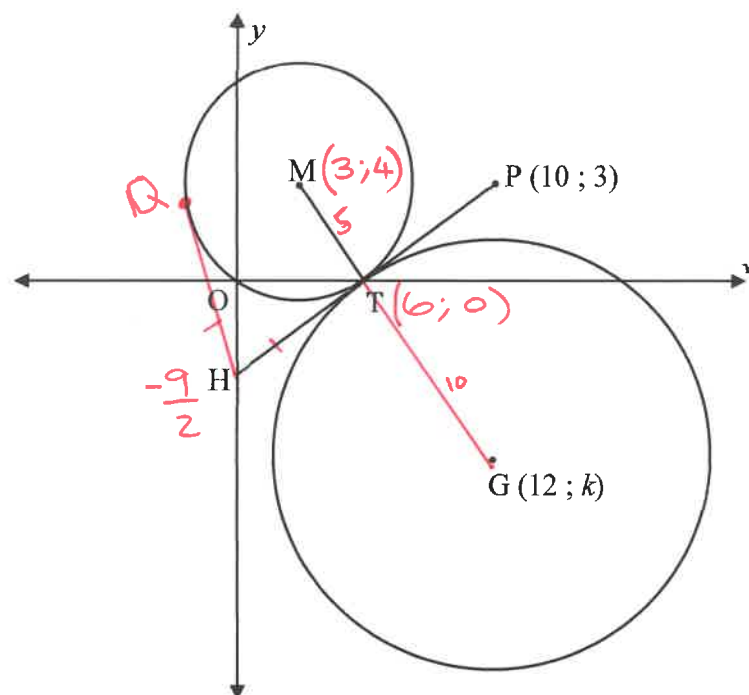
QUESTION 4



	Solution	Marks
4.1	$Rm = mT$ Diags of parm $\therefore M\left(\frac{-4+16}{2}, \frac{-6+14}{2}\right)$ $m(6; 4)$	(2)
4.2	$m_{RT} = \frac{-6-14}{-4-16}$ $= 1$	(2)
4.3	$SM = mA$ $\frac{S}{R}$ Diags of parm $\therefore A(18; -8)$	(3)

	Solution	Marks
4.4	STAR is a parm (given) $m_{SA} = \frac{16 - (-8)}{-6 - 18} \checkmark$ $m_{SA} = -1$ $\therefore m_{SA} \times m_{RT} = -1 \checkmark$ SA $\perp$ RT $\therefore$ STAR is a parm with $\perp$ diags.	(3)
4.5	$m_{SA} = -1$ $\therefore M\hat{H}x = 135 \checkmark m \checkmark$ $\therefore M\hat{H}K = 45 \checkmark$ LS on str line	(3)
4.6	N(-6; 4) $\therefore MN = 12 \text{ units} \checkmark \checkmark$	(2)
		[15]

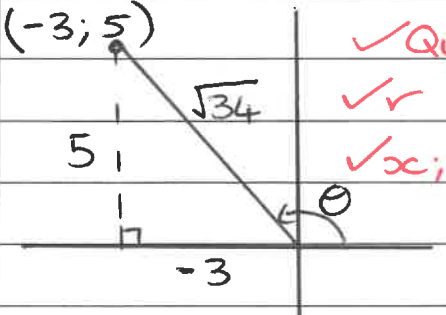
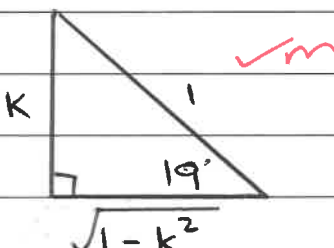
QUESTION 5



	Solution	Marks
5.1	$M(3; 4)$	(2)
5.2	$(x-3)^2 + (0-4)^2 = 25 \quad \checkmark y=0$ $x^2 - 6x + 9 - 16 = 0$ $x^2 - 6x - 7 = 0 \quad \checkmark$ $x(x-7) = 0$ $x=0 \quad \checkmark \quad \text{or} \quad x=7$ $\text{N/A} \quad \checkmark \quad \text{must be coordinate}$ $\therefore T(6; 0)$	(4)
5.3	$m_{MT} = \frac{4-0}{3-6} \quad \checkmark$ $= -\frac{4}{3} \quad \checkmark$	(2)

	Solution	Marks
5.4	$m_{HP} = \frac{3}{4} \checkmark \quad (\text{rad} \perp \text{tan}) \checkmark R$ $\therefore y - 3 = \frac{3}{4} (x - 10) \checkmark \quad y - y_1 = m(x - x_1)$ $y = \frac{3}{4}x - \frac{9}{2} \checkmark CA$	(4)
5.5	$H \left(0; -\frac{9}{2} \right) \checkmark$ $HT = HQ \quad \checkmark R \text{ Tangent from same point}$ $HT = \sqrt{(0 - 6)^2 + \left(-\frac{9}{2} - 0\right)^2} \checkmark$ $HT = 7,5 \text{ units}$ $\therefore HQ = 7,5 \text{ units} \checkmark$	(4)
5.6	$\frac{k - 4}{12 - 3} = -\frac{4}{3} \checkmark \quad \checkmark m_{MG} = -\frac{4}{3}$ $k - 4 = -12$ $k = -8 \checkmark$	(4)
5.7	$(x - 12)^2 + (y + 8)^2 = 100 \checkmark \quad \checkmark (10)^2$	(2)
		[22]

QUESTION 6

	Solution	Marks
6.1	$(-3; 5)$  $\checkmark \text{Quad}$ $r^2 = 5^2 + (-3)^2$ $\checkmark r$ $r^2 = 34$ $\checkmark x; y$ $r = \sqrt{34}$ $\cos 2\theta$ $= 2\cos^2 \theta - 1$ $\checkmark$ $\text{Can use any } \cos 2\theta$ $= 2\left(\frac{-3}{\sqrt{34}}\right)^2 - 1$ $= 2\left(\frac{9}{34}\right) - 1$ $= -\frac{8}{17} \checkmark$	(6)
6.2	$\sin 38 = 2 \sin 19 \cos 19 \checkmark$  $\checkmark m$ $= 2k \cdot \sqrt{1-k^2} \checkmark$	(3)

Alternative 5.6.

$$MG = 15$$

$$\sqrt{(12-3)^2 + (k-4)^2} = 15 \quad \checkmark \checkmark$$

$$81 + k^2 - 8k + 16 = 225$$

$$k^2 - 8k - 128 = 0$$

$$(k - 16)(k + 8) = 0 \quad \checkmark$$

$$k = 16$$

N/A

or

$$k = -8 \quad \checkmark$$

	Solution	Marks
6.3.1	$\frac{\sin(180-\theta) \cdot \tan(720+\theta) \cdot \cos(-\theta)}{-\cos(90+\theta) \cdot \tan(\theta-180)}$ $= \frac{\overset{\checkmark}\sin\theta \cdot \overset{\checkmark}\tan\theta \cdot \overset{\checkmark}\cos\theta}{-(-\overset{\checkmark}\sin\theta) \cdot \overset{\checkmark}\tan\theta}$ $= \cos\theta \checkmark$	(6)
6.3.2	$\frac{\cos 34^\circ \cos 10^\circ + \sin 34^\circ \sin 10^\circ}{\sin 12^\circ \cos 12^\circ}$ $= \frac{\cos(34-10)}{\sin 12^\circ \cos 12^\circ}$ $= \frac{\cos 24^\circ \checkmark}{\sin 12^\circ \cdot \cos 12^\circ}$ $= \frac{2 \cos 24^\circ \checkmark m}{2 \sin 12^\circ \cdot \cos 12^\circ}$ $= \frac{2 \cos 24^\circ}{\sin 24^\circ \checkmark}$ $= \frac{2}{\tan 24^\circ \checkmark}$	(4)

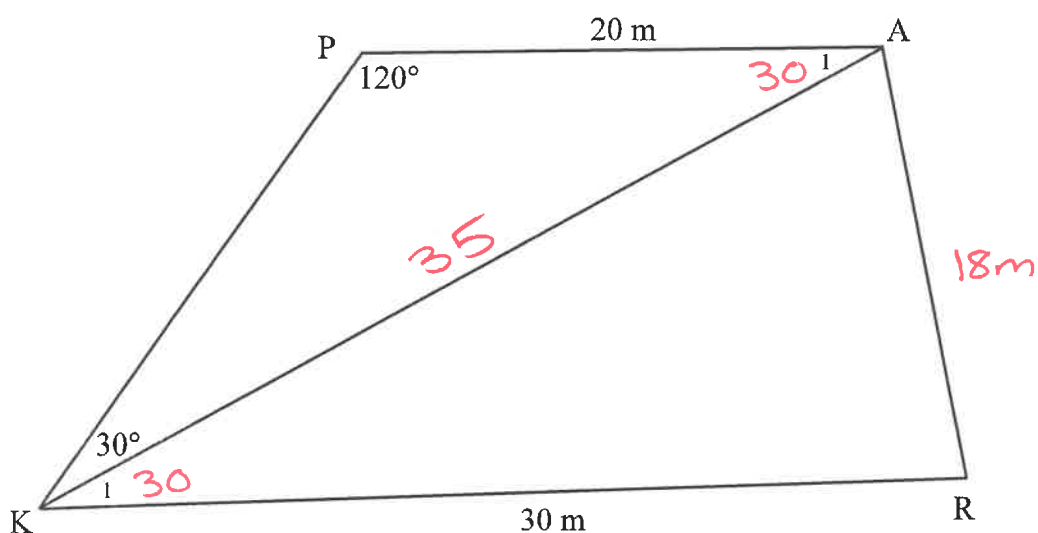
	Solution	Marks
6.4	$\text{LHS} = \frac{\sin 4x \cos 2x - 2 \cos 4x \sin x \cos x}{\tan 2x}$ $= \frac{\sin 4x \cos 2x - \cos 4x \sin 2x}{\tan 2x}$ $= \frac{\sin (4x - 2x)}{\tan 2x}$ $= \sin 2x \div \frac{\sin 2x}{\cos 2x}$ $= \sin 2x \times \frac{\cos 2x}{\sin 2x}$ $= \cos 2x$ $= \cos^2 x - \sin^2 x = \text{RHS}$ $\therefore \text{LHS} = \text{RHS}$	(5)
6.5	$\sin 3x = -5 \cos 3x$ $\tan 3x = -5$ $\text{ref } L = 78,69$ $3x = 101,31 + k \cdot 360$ $x = 33,77 + k \cdot 120 ; k \in \mathbb{Z}$ or $3x = 281,31 + k \cdot 360$ $x = 93,77 + k \cdot 120 ; k \in \mathbb{Z}$ $\therefore x \in \{33,77^\circ ; 93,77^\circ ; 153,77^\circ\}$	(6)
		[30]

QUESTION 7

✓ Sin
✓ period change
✓ shift ↑ 3.

7.1		(3)
7.2	$\frac{1}{2}$ ✓	(1)
7.3	$y \in [-1; \infty)$ ✓ or $y \geq -1$	(1)
7.4	360° ✓	(1)
7.5	$h(x) = 2 \cos(x - 30^\circ) - 4$ ✓	(2)
7.6	$2 \sin x \cos x + 3 > 3$ $\sin(2x) > 0$ $0 < x < 90^\circ$ ✓ ✓	(2)
		[10]

QUESTION 8

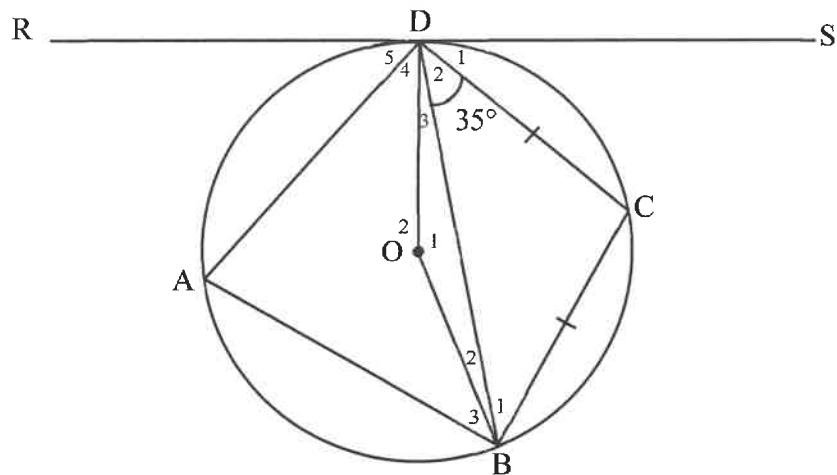


	Solution	Marks
8.1	$\frac{KA}{\sin \hat{P}} = \frac{AP}{\sin \hat{P}KA}$ $\frac{KA}{\sin 120} = \frac{20}{\sin 30} \quad \checkmark \text{sin rule}$ $KA \approx 35m \quad \checkmark \text{values}$	(3)
8.2	$\hat{A}_1 = 30^\circ \quad \checkmark \text{S/R } \angle\text{s in } \Delta$ $\therefore \hat{K}_1 = 30^\circ \quad \checkmark \text{S/R Alt } \angle\text{s ; PA} \parallel \text{KR}$ $AR^2 = KA^2 + KR^2 - 2 \cdot KA \cdot KR \cdot \cos \hat{K}_1$ $AR^2 = 35^2 + 30^2 - 2(35)(30) \cdot \cos 30^\circ \quad \checkmark \text{cos rule}$ $AR \approx 18m \quad \checkmark$	(4)

QUESTION 9

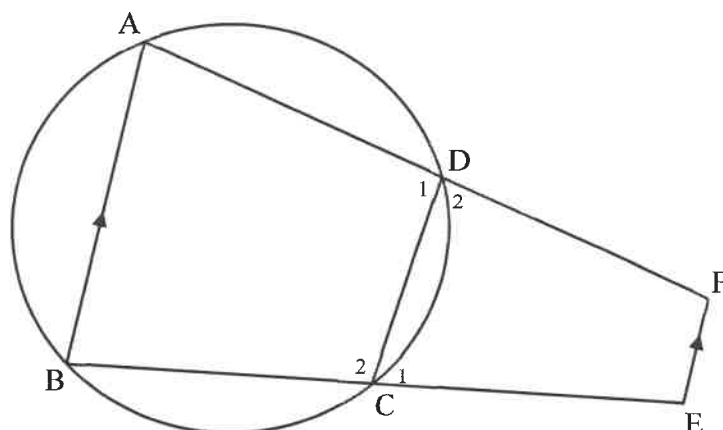
	Solution	Marks
9.1	$\hat{O}_1 = 2\hat{A}$ ✓ $\angle$ at centre = 2 x $\angle$ at circ $\hat{O}_2 = 2\hat{C}$ ✓ $\angle$ at centre = 2 x $\angle$ at circ $\hat{O}_1 + \hat{O}_2 = 360$ ✓ $\angle$s round a point $2\hat{A} + 2\hat{C} = 360$ ✓ $\therefore \hat{A} + \hat{C} = 180$	
		(5)

9.2



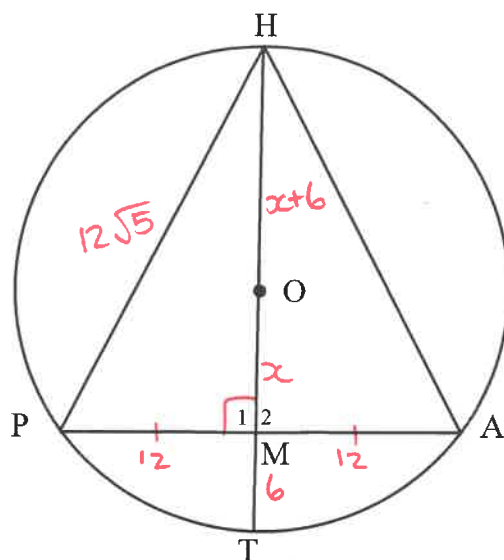
	Solution	Marks
9.2.1	$\hat{B}_1 = 35$ ✓ $\angle$'s opp = sides ✓ $\therefore \hat{D}_1 = 35$ ✓ Tan Chord Thrm ✓	(3)
9.2.2	$\hat{C} = 110$ ✓ $\angle$'s in a Δ ✓	(2)
9.2.3	$\hat{A} = 70$ ✓ opp $\angle$'s of cyclic quad ✓	(2)
9.2.4	$\hat{O}_1 = 140$ ✓ $\angle$ at centre = 2x $\angle$ at circ ✓	(2)
9.2.5	$\angle ODR = 90$ ✓ rad $\perp$ tan ✓	(2)
		[16]

QUESTION 10



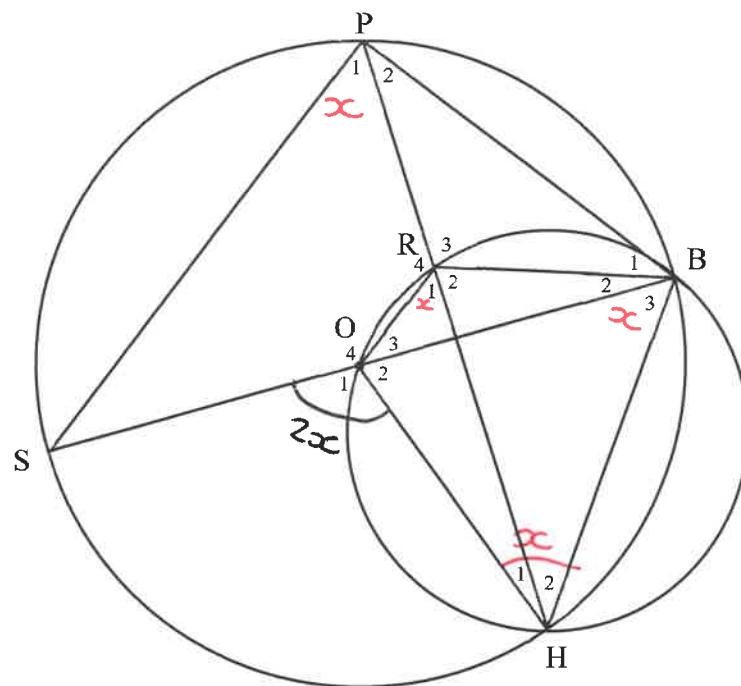
	Solution	Marks
	$\hat{C}_1 = \hat{A}$ ✓ Ext $\angle$ of cyclic quad	
	$\hat{A} + \hat{F} = 180$ ✓ Co-int $\angle$ s; $AB \parallel FE$	
	$\therefore \hat{C}_1 + \hat{F} = 180$ ✓	
	$\therefore$ CDFE is cyclic quad	
	Converse opp $\angle$ s of cyclic quad ✓ R	
	OR	
	Opp $\angle$ s of quad supp	
		(4)
		[4]

QUESTION 11



	Solution	Marks
11.1	$HO = OT = x + 6$ radii $HM = 2x + 6$ ✓✓	(2)
11.2	$\hat{m}_1 = 90$ ✓ line from centre to midpoint chord $HM^2 = PH^2 - PM^2$ Pythag ✓ $(2x + 6)^2 = (12\sqrt{5})^2 - (12)^2$ $4x^2 + 24x + 36 = 720 - 144$ $4x^2 + 24x - 540 = 0$ $x^2 + 6x - 135 = 0$ ✓ $(x + 15)(x - 9) = 0$ $x = -15$ or $x = 9$ ✓ N/A. ✓ $\therefore r = 15$ ✓ $\therefore \text{Area} = \pi (15)^2$ $= 706,86 \text{ units}^2$ ✓	(7)
		[9]

QUESTION 12



	Solution	Marks
12.1	$\hat{P}_1 = x$ ✓ $\angle$ at centre = $2 \times \angle$ at circ $\hat{B}_3 = x$ ✓ $\angle$'s in same segment $\hat{O}HB = x$ ✓ $\angle$'s opp = radii $\hat{R}_1 = x$ ✓ $\angle$'s in same segment	(4)
12.2	$\hat{O}_2 = 180 - 2x$ ✓ $\angle$'s on str line $\therefore \hat{R}_2 = 180 - 2x$ ✓ $\angle$'s in same segment ✓	(3)

	Solution	Marks
12.3	$\hat{S}\hat{P}B = 90^\circ$ ✓ $\angle$ in semi $\odot$ $\therefore \hat{P}_2 = 90 - x$ ✓ $\hat{R}_2 = \hat{P}_2 + \hat{B}_1$ ✓ Ext $\angle$ of $\triangle$ $\therefore 180 - 2x = 90 - x + \hat{B}_1$ $\therefore \hat{B}_1 = 90 - x$ ✓ $\therefore \hat{P}_2 = \hat{B}_1$ ✓ $\therefore RP = RB$ ✓ Sides opp = $\angle$'s	
		(6)
		[12]